

UCHEMADU NWACHUKWU

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Professional Summary

Clinical data analytics and research professional with 8+ years in critical care, clinical trial management, and health informatics. Skilled in **R programming, Shiny app development, tidyverse, data visualization (ggplot2, plotly), ETL pipeline design, and OMOP Common Data Model (DCM) implementation**. Experienced in **clinical trial workflows, GCP compliance, electronic CRFs, and open-source tool development**. Excellent at combining clinical expertise with advanced analytics to drive evidence-based decision-making, optimize data workflows, and improve patient outcomes.

Core Competencies

- **R Programming & Shiny Development:** R package development, Shiny dashboard deployment, production-ready analytics tools.
- **Clinical Data Management & OMOP:** ETL pipeline development, OMOP DCM mapping, CRFs/eCRFs, REDCap, patient recruitment, data cleaning.
- **Data Visualization & Reporting:** ggplot2, plotly, patchwork, interactive dashboards, automated reporting.
- **Predictive Analytics & Modeling:** Regression analysis, machine learning, risk prediction, splines.
- **Regulatory Compliance & GCP:** SOP implementation, audit readiness, ethical research conduct.
- **Software Development Practices:** Git version control, code review, documentation, testing, workflow optimization.
- **Project & Team Leadership:** Multidisciplinary collaboration, training, workflow modernization, knowledge transfer.

Education

- **MSc Applied Clinical Data Analytics (First Class Honours)** – University of Galway, Ireland, 2025
- **NFQ Level 9 Minor Award, Data Analytics for Cancer Real World Data Research** – University of Limerick, 2025
- **Master of Health Planning & Management (MHPM, Upper Second Class)** – University of Benin, Nigeria, 2018
- **Postgraduate Diploma in Education (PGDE, Upper Second Class)** – University of Benin, Nigeria, 2017
- **Bachelor of Nursing Science (BNSc)** – University of Nigeria, Nsukka, 2012

Certifications

- Good Clinical Practice (GCP), NIDA Clinical Trials Network
- Systematic Reviews, Evidence Synthesis and Cochrane Ireland, 2025
- Healthcare Data Management & Information Systems, Northeastern University, 2025
- Cybersecurity in Healthcare, Erasmus University Rotterdam, 2025
- Fundamentals of Critical Care Support (FCCS), Society of Critical Care Medicine, 2021
- Basic Life Support (BLS), American Heart Association, 2024
- Mechanical Ventilation for COVID-19, Harvard Medical School, 2022

Professional Membership

- Irish Computer Society (ICS)

Professional Experience

Researcher and Data Manager – Magnate Analytics Consulting | 2025 – Present

- Clean, validate, and analyze datasets using **R and Python**, supporting healthcare and research projects.
- Develop **R packages** and **Shiny dashboards** for interactive data visualization and workflow automation.
- Collaborate with multidisciplinary teams to interpret data, provide actionable insights, and improve analytics adoption.
- Support **audit, quality assurance, and data governance** initiatives aligned with GCP and regulatory standards.
- Deliver postgraduate lectures on **data visualization, predictive modelling, OMOP mapping, and R programming best practices**.

Assistant Chief Nursing Officer (ICU & Clinical Research Lead) – Irrua Specialist Teaching Hospital, Nigeria | 2016 – 2024

- Led ICU teams and managed **clinical trials**, including Lassa fever and COVID-19 studies.
- Oversaw patient recruitment, informed consent, CRFs/eCRFs, and **protocol compliance**.
- Conducted **data cleaning, ETL, statistical analysis, and evidence synthesis** for epidemiological and clinical studies.
- Implemented **healthcare analytics workflows** for workforce planning, patient safety, and quality improvement.
- Trained and supervised junior staff in **critical care, research methodology, and clinical data analytics**.

ICU Nurse – Memfys Hospital for Neurosurgery | 2014 – 2016

- Managed neurocritical and post-surgical patients, including **mechanical ventilation and ABG interpretation**.
- Mentored ICU staff and nursing students in clinical protocols and patient monitoring.

Technical Skills

- **Programming & Analytics:** R (tidyverse, ggplot2, plotly, Shiny), Python, predictive modeling, statistical analysis
- **Data Management & ETL:** REDCap, CRFs/eCRFs, ETL pipeline development, **OMOP DCM mapping**, workflow automation
- **Visualization & Reporting:** Shiny dashboards, patchwork, interactive dashboards, automated reporting
- **Software Development Tools:** Git, version control, code review, package documentation/testing

Clinical Skills

- Advanced ventilator management, sedation, and critical care monitoring
- Resuscitation, trauma, and post-surgical ICU care
- Patient-centered care and adverse event management
- Clinical sample handling and biospecimen management

Hobbies & Interests

- Data visualization challenges, open-source contributions
- Public speaking and science communication
- Writing and publishing health-related books
- Community service initiatives

References

- **Dr. Conor Judge** – Co-Director & Senior Lecturer, Applied Clinical Data Analytics, University of Galway | conor.judge@universityofgalway.ie
- **Dr. Finn Krewer** – Lecturer, Applied Clinical Data Analytics, University of Galway | finn.krewer@universityofgalway.ie
- **Mrs. Maries Esezobor** – Deputy Director of Nursing, Irrua Specialist Teaching Hospital | mariesesezobor@gmail.com